

Abhi Chandra Yayi

abhichandrayayi@gmail.com | +1 (602) 727-5348 | [linkedin.com/in/abhi-chandra-yayi/](https://www.linkedin.com/in/abhi-chandra-yayi/) | abhiyayi04.github.io/Portfolio Website/

EDUCATION

B.S. Information Technology

August 2022 – May 2026

Arizona State University (GPA: 3.98 | Dean's List: 7 semesters)

Tempe, Arizona

- Coursework: Object-Oriented Programming, Database Management, Web development, Project Management

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Java, JavaScript, HTML, CSS, Bash, Go

Libraries and Frameworks: Django, Flask, RESTful APIs, React, Scikit-Learn, PyTorch, MongoDB, Pandas

Tools: Linux, Git, Docker, Kubernetes, Alteryx, Snowflake, Tableau, AWS (ECS, Lambda, SQS, Sagemaker), Claude

AI/ML: Neural Networks, Multimodal Transformers, LLM fine-tuning, Evaluation & Metrics, RAG, MLOps, MCP

EXPERIENCE

Data Analyst Intern

June 2025 – July 2025

Food Forest AI

Philadelphia, PA

- Automated and standardized data cleaning for company records using **Python** and **regex**, reducing manual processing time by **80%** and improving data accuracy for downstream enrichment.
- Validated and audited datasets using **SQL** by detecting duplicates, null values, and schema inconsistencies, ensuring data readiness and improving ingestion reliability by **40%**.
- Developed and evaluated a **text classification NLP** pipeline using **embeddings-based** similarity to categorize company descriptions to automate company categorization, reducing manual effort.

Solutions Architect

September 2024 – May 2026

Arizona State University

Tempe, AZ

- Architected technical solutions to **20–30** peers weekly across data structures, debugging, and distributed systems, driving **collaborative problem-solving**, peer **code reviews**, and improved **software development** practices.
- Led coordination between faculty and a team of **10+** Learning Assistants by streamlining technical **communication**, documenting instructional workflows, and improving peer support and **operational efficiency**.

PROJECTS

Agentic Research Assistant (LangGraph, LangChain, FastAPI, Qdrant, MySQL, Docker, LangSmith)

- Developed an **agentic RAG** platform that ingests files, enabling users to ask complex analytical questions and receive cited, multi-step answers generated through tool-using **AI agents**.
- Designed **MCP-connected LangGraph** workflows for vector retrieval, SQL querying, Python-based analysis, citation validation, and report generation using **FastAPI**, **Qdrant**, **MySQL**, and **Docker**.
- Reduced simulated manual research time by **~80%** across benchmark business-analysis tasks while using **LangSmith** to measure retrieval quality, faithfulness, latency, and citation accuracy.

Movie Recommendation Engine (XGBoost, FastAPI, MLflow, Prometheus, Grafana)

- Developed a personalized movie recommendation platform using **collaborative filtering** and **XGBoost** ranking, generating real-time recommendations through a **FastAPI** service with cold-start support for new users.
- Designed an end-to-end **MLOps** pipeline using **Scikit-Learn**, **XGBoost**, **MLflow**, **Redis**, **PostgreSQL**, GitHub Actions, and Docker, enabling automated model training, experiment tracking, deployment, and monitoring.
- Improved recommendation quality by **28% NDCG@10** over a popularity-based baseline and implemented A/B testing to evaluate CTR and conversion, with model versioning and inference audit logging.

Smart Finance Tracker (Django, Python, HTML, CSS, MySQL, Git, AWS, OpenAI)

- Engineered a full-stack smart bookkeeping system that enables users to track transactions, categorize spending, import financial data via CSV, and visualize financial insights through interactive dashboards.
- Designed a scalable 3-tier architecture using **Django templates** and **bootstrap** (presentation layer), **Django** (application layer), **MySQL** on **AWS RDS** (data layer), and deployed it using **AWS EC2 + VPC**.
- Integrated an AI expense-categorization feature using structured **prompt engineering** with an OpenAI **API call**, automatically classifying transactions and reducing manual labeling time by about **65%**.

CERTIFICATIONS

- Back-End Developer Professional Certificate – Meta (Issued: November, 2025)*
- Machine Learning Specialization – DeepLearning.AI & Stanford Online (Issued: August, 2025)*